Revision-Note Review Worksheet for Expert Rater

This booklet contains one student revision note to review. It is a take-home revision note prepared by a teacher for a JSS1 Mathematics class on Three dimensional figures: properties of cubes, cuboids, pyramids and cones. Judge how well it covers the intended objectives and curriculum, follows the teacher's lesson plan, stays on topic and is conceptually correct, and is clear and useful for the students.

Lesson and revision-note context

Subject	Mathematics
Grade	JSS1
Term	Third Term
Week	4
Topic	Three Dimensional Shapes
Subtopic	Three dimensional figures: properties of cubes, cuboids, pyramids and cones
Theme	—
Curriculum objectives	• Identify the properties of cube and cuboids • Identify the properties of pyramids and cones
Previous knowledge	• Students have been taught the differences between cube and cuboid.
Lesson sequence	• Identify the properties of cube and cuboids — the teacher writes the properties of cubes and cuboids on the board; the student take notes in their notebooks • Identify the properties of pyramids and cones — the teacher draws the pyramids and the cone on the board and writes their properties; the student take them down, as in Step II
Instructional materials	• Shapes of cubes, cuboid, Pyramid, cone, cylinder and spheres.
Evaluation	• How many vertices, has a cuboid? • How many edges has a cube? • Draw a cube and cuboid then label. • Draw an hexagonal base pyramid and list the properties. • How many vertices has a tetrahedron.

Lesson objectives

1.	Identify the properties of cube and cuboids
2.	Identify the properties of pyramids and cones

Read the student revision note on the following pages, then complete the rating sheet at the end of this booklet.

STUDENT REVISION NOTE

Topic: Three dimensional figures: properties of cubes, cuboids, pyramids and cones

Class: JSS1 **Subject:** Mathematics **Week:** 4

Section 1: Properties of Cubes and Cuboids

Three-dimensional shapes are shapes that have length, breadth and height. Because they have all three of these, they are called **solid shapes**. They can be regular or irregular. A cube of sugar, a can drink and a matchbox are regular shapes; a stone or a tree is an irregular shape.

Two important solids you will meet are the **cube** and the **cuboid**. Before we list their properties, make sure you know three words that we use to describe any solid shape:

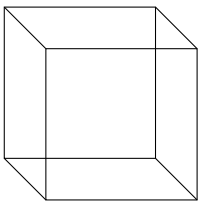
- A **face** (or surface) is one of the flat surfaces of the shape. A face may be flat or curved.
- An **edge** is a line where two faces meet. An edge may be straight or curved.
- A **vertex** is a point corner where edges meet. (More than one vertex are called **vertices**.)

Properties of a cube

A cube is a solid in which **all six faces are squares** of the same size. Look at fig-01 — a cube drawn as a see-through solid, with two equal squares joined corner to corner. The properties of a cube are:

1. It is a solid state.
2. It has **6 square faces**.
3. It has **12 edges**.
4. It has **8 vertices**.

Visual



A cube drawn as a see-through solid: two equal squares, the nearer one offset down and to the right of the farther one, with the four pairs of corresponding corners joined.

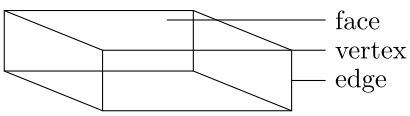
Properties of a cuboid

A cuboid is a solid like the one in fig-02, drawn wide and shallow. Its faces are **rectangles**, not all of them equal. The properties of a cuboid are:

1. It has **6 faces**.
2. It has **12 edges**.
3. It has **8 vertices**.

In fig-02 the leader lines name three parts for you: the top **face**, a **vertex** (the nearer top-right corner, where edges meet) and an **edge** (the nearer right-hand vertical line where two faces meet).

Visual



A cuboid drawn the same way, wide and shallow, with three leader lines to the right naming the top face, the nearer top-right corner and the nearer right-hand vertical edge.

Worked example. How many edges does a cube have, and what shape are its faces?

A cube has **12 edges** and its faces are all **squares**. Because all six faces are squares of the same size, the cube is the solid whose faces are all equal squares.

Practice

1. How many vertices does a cuboid have?
2. How many edges does a cube have?
3. A cube and a cuboid both have the same number of faces. How many faces does each one have?
4. In your own words, what is the difference between the faces of a cube and the faces of a cuboid?

Section 2: Properties of Pyramids and Cones

You already know the difference between a cube and a cuboid. Now we meet two more solids: pyramids and cones.

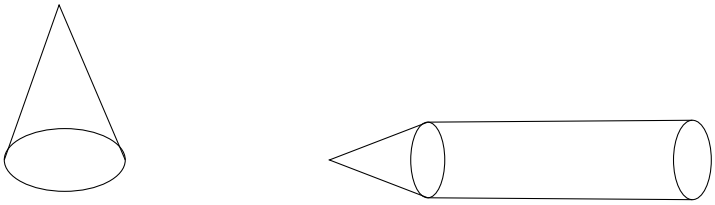
A pyramid is named according to the shape of its base. Look at fig-03: a pyramid with a triangular base is called a triangular based pyramid. Because it has four triangular faces, it is also called a tetrahedron (from the Greek for "four faces"). fig-04 shows a square based pyramid, fig-05 a pentagonal based pyramid, and fig-06 a rectangular based pyramid. In every pyramid, the base is a flat polygon and the slanted faces meet at a single point at the top called the apex.

Count the faces, edges and vertices of each pyramid carefully, and you find this pattern:

Shape	Faces	Edges	Vertices
Triangular based pyramid	4	6	4
Square based pyramid	5	8	5
Pentagonal based pyramid	6	10	6
Rectangular based pyramid	5	8	5

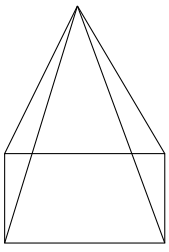
A cone (fig-07) is a different solid, with a round base and a pointed top. It has two faces — one curved face and one flat circular face. It has one edge (the circular rim where the two faces meet) and one vertex (the pointed apex).

Visual



Two drawings side by side. On the left an upright cone: an elliptical base with the two slant sides running up from the ends of the ellipse to the apex. On the right a cone lying on its side, apex at the left, its two slant sides reaching a narrow upright ellipse whose top and bottom then continue as two parallel lines to a second narrow ellipse further right.

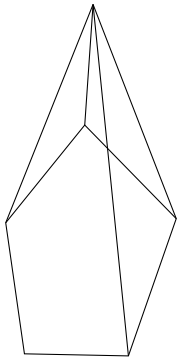
Visual



Rectangular based Pyramid

A rectangular based pyramid, drawn like the square one but on a base wider than it is tall.

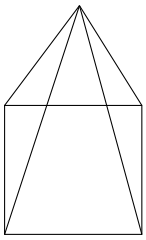
Visual



Pentagonal based Pyramid

A pentagonal based pyramid. The base is drawn as a pentagon standing on one side with its fifth vertex uppermost, and the apex above is joined to that vertex, to the two vertices either side of it, and to the lower-right vertex — four edges from the apex, the last of them crossing the face of the solid.

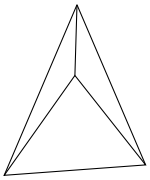
Visual



Square base Pyramid

A square base pyramid. The base is drawn as an upright rectangle and the apex above is joined to all four of its corners, so two edges run outside the rectangle and two cross it.

Visual



Triangular base Pyramid (Tetrahedron)

A triangular base pyramid. The base triangle is drawn in perspective, its third vertex raised into the middle of the figure, and the apex above is joined to all three base vertices.

Let's work out the properties of a square based pyramid by looking at fig-04.

Step 1 — Faces. The base is a square, and four triangular faces slant up from the four sides of the square to the apex. So there are 5 faces.

Step 2 — Edges. The base has 4 edges, and each corner of the base is joined to the apex by one more edge, giving 4 more. So there are $4 + 4 = 8$ edges.

Step 3 — Vertices. There are the 4 corners of the base plus the apex at the top. So there are 5 vertices.

So a square based pyramid has 5 faces, 8 edges and 5 vertices, which matches the table.

Practice

1. How many vertices has a tetrahedron?
2. A pentagonal based pyramid — how many faces, edges and vertices does it have?
3. Write down the number of faces, edges and vertices of a cone.

Check your work

Attempt every question before you read this page.

Properties of Cubes and Cuboids

1. 8 vertices
2. 12 edges
3. 6 faces
4. A cube has 6 square faces, all of the same size; a cuboid has 6 faces that are rectangles, not all equal.

Properties of Pyramids and Cones

1. 4

   A tetrahedron is a triangular based pyramid. From the table, a triangular based pyramid has 4 vertices.
2. 6 faces, 10 edges and 6 vertices.

   From the table, a pentagonal based pyramid has 6 faces, 10 edges and 6 vertices.
3. A cone has 2 faces (one curved, one flat), 1 edge and 1 vertex.

Rating sheet

Mark one score per criterion, then use the notes section for strengths, concerns, and recommendations. The lesson objectives are on the first page for reference.

Revision note ID	cand-e4728b0d	**Date**	
Subtopic	Three dimensional figures: properties of cubes, cuboids, pyramids and cones	**Rater**	

> The reference sheet includes the teacher's evaluation questions. A pack that reuses them, reworded or verbatim, may be credited for matching how the lesson was assessed — but reuse on its own does not establish taught-lesson alignment. Rate alignment on the whole of it: the objectives covered, the order taught, what the activities had the students do, the method, and the expected depth.

Criterion (select one score per row)	**1** Poor	**2** Weak	**3** Adequate	**4** Good	**5** Excellent
Objective and curriculum alignment Does the revision note cover the intended lesson objectives and curriculum target?	○	○	○	○	○
Taught-lesson alignment Does it reflect the taught lesson — its sequence, emphasis, activities, and expected depth?	○	○	○	○	○
Scope and conceptual correctness Does it stay within the expected lesson boundary and avoid conceptual/mathematical errors?	○	○	○	○	○
Student-facing clarity and usefulness Can JSS1 learners understand and use it meaningfully as take-home review material?	○	○	○	○	○

Strengths	Concerns	Recommendation

Flag any problems (optional)

Tick all that apply. These are separate from the scores above — leave blank if none apply.

☐	**Misses a required objective** — Skips something the lesson was meant to teach.
☐	**Goes beyond this lesson** — Includes material outside this lesson's scope (e.g. next week's topic).
☐	**Wrong or misleading content** — Contains a mathematical or factual mistake.
☐	**Too hard for JSS1 to follow** — Likely confusing for the students.
☐	**Other:** _____